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・根号を含む数列の極限

(例) 次の極限を求めよ。

point
根号を含む不定形 $\infty-\infty$ の場合
有理化

(1) $\displaystyle\lim_{n\to\infty}\frac{n}{\sqrt{n^2+n}+n}$ $\quad\frac{\infty}{\infty}$ $\quad\sqrt{n^2+n}=\sqrt{n^2(1+\frac{1}{n})}$

$\displaystyle=\lim_{n\to\infty}\frac{1}{\sqrt{1+\frac{1}{n}}+1}$ $\quad\frac{1}{\sqrt{1+0}+1}$ $\quad=|n|\sqrt{1+\frac{1}{n}}\ (\because \sqrt{A^2}=|A|)$

$=n\sqrt{1+\frac{1}{n}}\ (\because n>0)$

$=\frac{1}{2}$ //

(2) $\displaystyle\lim_{n\to\infty}\frac{1}{n-\sqrt{n^2+4n}}$ $\quad\frac{1}{\infty-\infty}$

$\displaystyle=\lim_{n\to\infty}\frac{1}{n-\sqrt{n^2+4n}}\cdot\frac{n+\sqrt{n^2+4n}}{n+\sqrt{n^2+4n}}$

$\displaystyle=\lim_{n\to\infty}\frac{n+\sqrt{n^2+4n}}{n^2-(\sqrt{n^2+4n})^2}$

$\displaystyle=\lim_{n\to\infty}\frac{n+\sqrt{n^2+4n}}{-4n}$ $\quad\frac{\infty}{-\infty}$

$\displaystyle=\lim_{n\to\infty}\frac{1+\sqrt{1+\frac{4}{n}}}{-4}$ $\quad\frac{1+\sqrt{1+0}}{-4}$

$=-\frac{1}{2}$ //

(3) $\displaystyle\lim_{n\to\infty}(\sqrt{n^2+1}-n)$ $\quad\infty-\infty$

$\displaystyle=\lim_{n\to\infty}(\sqrt{n^2+1}-n)\frac{\sqrt{n^2+1}+n}{\sqrt{n^2+1}+n}$

$\displaystyle=\lim_{n\to\infty}\frac{(\sqrt{n^2+1})^2-n^2}{\sqrt{n^2+1}+n}$

$\displaystyle=\lim_{n\to\infty}\frac{1}{\sqrt{n^2+1}+n}$

$=0$ //

(4) $\displaystyle\lim_{n\to\infty}\sqrt{n}(\sqrt{n+2}-\sqrt{n+1})$ $\quad\infty(\infty-\infty)$

$\displaystyle=\lim_{n\to\infty}\sqrt{n}(\sqrt{n+2}-\sqrt{n+1})\cdot\frac{\sqrt{n+2}+\sqrt{n+1}}{\sqrt{n+2}+\sqrt{n+1}}$

$\displaystyle=\lim_{n\to\infty}\frac{\sqrt{n}\{(\sqrt{n+2})^2-(\sqrt{n+1})^2\}}{\sqrt{n+2}+\sqrt{n+1}}$

$\displaystyle=\lim_{n\to\infty}\frac{\sqrt{n}}{\sqrt{n+2}+\sqrt{n+1}}$ $\quad\frac{\infty}{\infty}$

$\displaystyle=\lim_{n\to\infty}\frac{1}{\sqrt{1+\frac{2}{n}}+\sqrt{1+\frac{1}{n}}}$ $\quad\frac{1}{\sqrt{1+0}+\sqrt{1+0}}$

$=\frac{1}{2}$ //

(5) $\displaystyle\lim_{n\to\infty}n^2(\sqrt[3]{n^3+1}-n)$ $\quad\infty\cdot(\infty-\infty)$ $\quad(a-b)(a^2+ab+b^2)=a^3-b^3$

$\displaystyle=\lim_{n\to\infty}n^2(\sqrt[3]{n^3+1}-n)\cdot\frac{(\sqrt[3]{n^3+1})^2+n\cdot\sqrt[3]{n^3+1}+n^2}{(\sqrt[3]{n^3+1})^2+n\cdot\sqrt[3]{n^3+1}+n^2}$

$\displaystyle=\lim_{n\to\infty}\frac{n^2\{(\sqrt[3]{n^3+1})^3-n^3\}}{(\sqrt[3]{n^3+1})^2+n\cdot\sqrt[3]{n^3+1}+n^2}$

$\displaystyle=\lim_{n\to\infty}\frac{n^2}{(\sqrt[3]{n^3+1})^2+n\cdot\sqrt[3]{n^3+1}+n^2}$

$\displaystyle=\lim_{n\to\infty}\frac{1}{(\sqrt[3]{1+\frac{1}{n^3}})^2+1\cdot\sqrt[3]{1+\frac{1}{n^3}}+1}$ $\quad\frac{1}{(\sqrt[3]{1+0})^2+1\cdot\sqrt[3]{1+0}+1}$

$=\frac{1}{3}$ //